Non-Abelian Information Loss under Homomorphic Binary Coarse-Graining of D_4: Fourier Reconstruction, Symmetric Dynamics, and Coherent Reference Recovery

D_4 群同态二值粗粒化中的非阿贝尔信息损失：群傅里叶重建、对称动力学与相干参考恢复

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Abstract

Binary readouts are often sufficient for reporting coarse outcomes but need not preserve the algebraic structure of the underlying process. We study this issue in the dihedral group D_4 of square symmetries (order eight), using its abelianization $D_4 \to \mathbb{Z}_2^2$ as a mathematically controlled model of homomorphic binary coarse-graining. The abelianized readout preserves three nontrivial one-dimensional characters but is blind to commutator information such as $[r, s] = r^2$. We give an exact finite-group Fourier reconstruction: three scalar character moments determine the abelian fiber weights, while a single 2×2 matrix in the unique two-dimensional irreducible representation recovers the four missing non-Abelian degrees of freedom. This yields a non-Abelian residual energy $\mathcal{E}_{\mathrm{NA}}$ and a precise information budget for successive $8 \to 4 \to 3$ coarse-grainings and context-rich refinements. For a two-rate D_4 random walk, representation theory produces the complete relaxation spectrum and the parameter-free binary relation $\gamma_{RS} = \gamma_R + \gamma_S$. Finally, in the standard two-dimensional quantum representation $R = -iY$ and $S = Z$, the commutator is $-I$: it is invisible as a target-system quantum channel but becomes observable as a relative phase when a coherent reference is retained. The framework is entirely within standard probability theory, finite-group representation theory, and standard quantum mechanics. Its purpose is not to propose a new quantum ontology, but to formalize when coarse binary descriptions erase conditional or path-sensitive information and how that information can be recovered by structured readout or coherent reference.

中文摘要。二值读出通常足以报告粗粒度结果，但未必保留底层过程的代数结构。本文以正方形对称群 D_4（八阶二面体群）为模型，用其阿贝尔化 $D_4 \to \mathbb{Z}_2^2$ 表示一种受控的“群同态二值粗粒化”。该粗粒化能够保留三个非平凡一维特征，却必然看不见 $[r, s] = r^2$ 一类交换子信息。本文给出精确的有限群傅里叶重建：三个标量特征矩确定四个阿贝尔纤维的概率和，而唯一二维不可约表示中的一个 2×2 矩阵恰好恢复其余四个非阿贝尔自由度。由此得到非阿贝尔残差能量 $\mathcal{E}_{\mathrm{NA}}$，并对连续的 $8 \to 4 \to 3$ 粗粒化和上下文保持细化给出精确信息预算。对于一个双速率 D_4 随机游走，表示论给出完整弛豫谱以及无额外自由参数的关系 $\gamma_{RS} = \gamma_R + \gamma_S$。最后，在标准二维量子表示 $R = -iY$、$S = Z$ 中，交换子为 $-I$：它在目标系统量子通道层完全不可见，但在保留相干参考时会转化为可观测相对相位。本文完全位于标准概率论、有限群表示论和标准量子力学内部，不提出新的量子本体；目标是严格刻画粗二值描述何时删除条件/路径敏感信息，以及如何通过结构化读出或相干参考恢复这些信息。

Keywords: dihedral group; D_4; binary coarse-graining; non-Abelian Fourier analysis; representation theory; hidden-state dynamics; quantum instruments; coherent reference; Hadamard test.

关键词：二面体群；D_4；二值粗粒化；非阿贝尔傅里叶分析；表示论；隐藏状态动力学；量子仪器；相干参考；Hadamard test（阿达玛测试）。

1 Introduction

A coarse observable is useful precisely because it discards detail. The same operation, however, can remove information that later becomes relevant for prediction, feedback, or path-sensitive control. This tension appears in many settings: hidden-state models, coarse-grained Markov dynamics, quantum instruments, and channel descriptions that average over conditional branches. The central question of this paper is deliberately narrow:

> *What algebraic information is lost when a non-Abelian finite-state process is compressed into homomorphic binary observables, and what is minimally required to recover it?*

We answer this question exactly for the symmetry group of a square,

$$D_4 = \langle r, s \mid r^4 = e,\ s^2 = e,\ srs = r^{-1} \rangle, \tag{1}$$

where r is a 90° rotation and s a reflection. We use the convention that D_4 has eight elements. The standard ingredients employed below—abelianization, finite-group Fourier analysis, random walks on groups, and coherent phase readout—are classical results in group theory and quantum information [1, 2, 3, 6]. The contribution is the integrated coarse-graining/recovery framework and the set of exact reconstruction, information-budget, dynamic, and operational consequences that follow from placing these ingredients in one model.

The results can be summarized in four statements. First, the natural abelianization is

$$D_4/[D_4, D_4] \cong \mathbb{Z}_2 \times \mathbb{Z}_2, \tag{2}$$

with kernel $\{e, r^2\}$. Thus each four-state coarse outcome hides one binary fiber coordinate. Second, the three nontrivial one-dimensional characters reconstruct all abelian information, while the unique two-dimensional irreducible representation carries exactly four additional linear degrees of freedom, sufficient to reconstruct the full eight-state distribution. Third, a symmetric two-rate random walk has a representation-theoretic relaxation spectrum whose one-dimensional part is directly accessible to three binary character readouts, while the remaining doubly degenerate modes live in the two-dimensional non-Abelian sector. Fourth, a standard quantum representation makes the commutator residue operationally visible only relative to a coherent reference: $[R, S] = -I$ is an identity channel on the target but a π phase shift on a control interferometer.

The scope is intentionally conservative. None of these facts modifies the Born rule, introduces hidden variables, or lies outside standard quantum mechanics. A binary readout that is *not* a homomorphism can, in principle, be designed to distinguish non-Abelian states directly. Our claims concern the specific and important class of homomorphic binary coarse-grainings and their combinations.

2 The D_4 coarse-graining hierarchy

2.1 Abelianization: the natural $8 \to 4$ map

The commutator subgroup of D_4 is

$$[D_4, D_4] = \{e, r^2\} \cong \mathbb{Z}_2. \tag{3}$$

Indeed, $[r, s] = rsr^{-1}s^{-1} = r^2$, and r^2 is central. Hence the quotient is the Klein four group.

Theorem 1 (Abelian binary quotient)**.** *The abelianization map*

$$\pi : D_4 \longrightarrow D_4/[D_4, D_4] \cong \mathbb{Z}_2^2 \tag{4}$$

is a two-to-one homomorphism. One convenient binary coding is

$$\{e, r^2\} \mapsto 00, \tag{5}$$
$$\{r, r^3\} \mapsto 10, \tag{6}$$
$$\{s, r^2 s\} \mapsto 01, \tag{7}$$
$$\{rs, r^3 s\} \mapsto 11. \tag{8}$$

Proof. In the abelianized group, $rs = sr$. Combining this with $srs = r^{-1}$ gives $r = r^{-1}$ and therefore $r^2 = e$ in the quotient. The independent images of r and s are both order two and commute, giving $\mathbb{Z}_2^2$. The kernel is $\{e, r^2\}$. □

This map preserves the parity of the rotation exponent and the parity of the reflection exponent, while deleting the distinction between g and gr^2 inside each fiber.

Corollary 2 (Path-order blindness of the abelian quotient)**.** *The quotient cannot distinguish noncommuting words that differ by the commutator fiber. In particular,*

$$rs \neq sr, \qquad \pi(rs) = \pi(sr) = 11, \tag{9}$$

and

$$r \neq r^3, \qquad \pi(r) = \pi(r^3) = 10. \tag{10}$$

The statement is structural, not statistical: no amount of repeated observation of $\pi(g)$ alone can distinguish members of the same fiber unless additional information is supplied.

2.2 A secondary $4 \to 3$ relation quotient

Let $V_4 = \mathbb{Z}_2^2$ and let $\tau(x, y) = (y, x)$ exchange the two binary coordinates. The orbit partition under $\langle\tau\rangle$ is

$$\{00\}, \qquad \{11\}, \qquad \{01, 10\}. \tag{11}$$

This gives a three-class relation space. It is an orbit quotient of the set V_4 rather than, in general, a quotient group. It is useful when an observation retains equality/inequality structure but discards the order of the two binary coordinates.

The resulting hierarchy is therefore

$$D_4 \xrightarrow{\text{abelianization}} \mathbb{Z}_2^2 \xrightarrow{\text{coordinate-swap orbits}} \{00, 11, D\}, \tag{12}$$

where $D = \{01, 10\}$.

3 Exact Fourier reconstruction of the lost non-Abelian information

3.1 Abelian fiber sums and fiber differences

Let $p(g)$ be an arbitrary probability distribution on D_4. Define the four coarse fiber weights

$$q_{00} = p_e + p_{r^2}, \tag{13}$$
$$q_{10} = p_r + p_{r^3}, \tag{14}$$
$$q_{01} = p_s + p_{r^2 s}, \tag{15}$$
$$q_{11} = p_{rs} + p_{r^3 s}, \tag{16}$$

with $\sum q_{ab} = 1$. These three independent degrees of freedom are exactly the information accessible through the three nontrivial one-dimensional characters of D_4.

Define also the four within-fiber differences

$$d_{00} = p_e - p_{r^2}, \tag{17}$$
$$d_{10} = p_r - p_{r^3}, \tag{18}$$
$$d_{01} = p_s - p_{r^2 s}, \tag{19}$$
$$d_{11} = p_{rs} - p_{r^3 s}. \tag{20}$$

These are invisible to the abelian quotient.

3.2 One-dimensional characters

Choose the three nontrivial real characters

$$\chi_R(r) = -1, \qquad \chi_R(s) = +1, \tag{21}$$
$$\chi_S(r) = +1, \qquad \chi_S(s) = -1, \tag{22}$$
$$\chi_{RS}(r) = -1, \qquad \chi_{RS}(s) = -1. \tag{23}$$

Together with normalization, the moments $\langle \chi_R \rangle$, $\langle \chi_S \rangle$, and $\langle \chi_{RS} \rangle$ are equivalent to the four q_{ab}. Explicitly,

$$q_{00} = \tfrac{1}{4}(1 + \langle \chi_R \rangle + \langle \chi_S \rangle + \langle \chi_{RS} \rangle), \tag{24}$$
$$q_{10} = \tfrac{1}{4}(1 - \langle \chi_R \rangle + \langle \chi_S \rangle - \langle \chi_{RS} \rangle), \tag{25}$$
$$q_{01} = \tfrac{1}{4}(1 + \langle \chi_R \rangle - \langle \chi_S \rangle - \langle \chi_{RS} \rangle), \tag{26}$$
$$q_{11} = \tfrac{1}{4}(1 - \langle \chi_R \rangle - \langle \chi_S \rangle + \langle \chi_{RS} \rangle). \tag{27}$$

3.3 The unique two-dimensional irreducible sector

Take the standard real two-dimensional representation

$$R = \rho_2(r) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad S = \rho_2(s) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \tag{28}$$

and set $X = RS$. Then $R^4 = S^2 = I$ and $SRS = R^{-1}$. The non-Abelian Fourier matrix is

$$M = \widehat{p}(\rho_2) = \sum_{g \in D_4} p(g)\rho_2(g). \tag{29}$$

Theorem 3 (Exact non-Abelian reconstruction)**.** *The matrix M has the orthogonal expansion*

$$\boxed{M = d_{00} I + d_{10} R + d_{01} S + d_{11} X.} \tag{30}$$

Moreover,

$$d_{00} = \tfrac{1}{2}\,\mathrm{Tr}(M), \tag{31}$$
$$d_{10} = \tfrac{1}{2}\,\mathrm{Tr}(R^{\mathsf{T}} M), \tag{32}$$
$$d_{01} = \tfrac{1}{2}\,\mathrm{Tr}(S^{\mathsf{T}} M), \tag{33}$$
$$d_{11} = \tfrac{1}{2}\,\mathrm{Tr}(X^{\mathsf{T}} M). \tag{34}$$

Therefore the three nontrivial character moments plus M reconstruct the full distribution $p(g)$ exactly.

Proof. The eight matrices are

$$I, R, -I, -R, S, RS, -S, -RS \tag{35}$$

for $e, r, r^2, r^3, s, rs, r^2s, r^3s$, respectively. Grouping opposite pairs yields Eq. (30). The basis $\{I, R, S, X\}$ is orthogonal under the Frobenius inner product $\langle A, B\rangle_F = \mathrm{Tr}(A^\mathsf{T} B)$ and every basis element has squared norm two, giving Eq. (34). Finally,

$$p_{g_+} = \frac{q+d}{2}, \qquad p_{g_-} = \frac{q-d}{2} \tag{36}$$

within each fiber. $\square$

This has an immediate dimension count. An eight-point probability distribution has seven independent real parameters. The three nontrivial one-dimensional moments contribute three. The 2×2 real matrix M contributes the remaining four. This matches the regular-representation decomposition

$$\mathbb{C}[D_4] \cong 1 \oplus \chi_R \oplus \chi_S \oplus \chi_{RS} \oplus 2\rho_2, \tag{37}$$

consistent with $4 \times 1^2 + 2^2 = 8$.

3.4 A non-Abelian residual energy

Define

$$\boxed{\mathcal{E}_{\mathrm{NA}} = \frac{1}{2}\|M\|_F^2 = d_{00}^2 + d_{10}^2 + d_{01}^2 + d_{11}^2.} \tag{38}$$

Then $\mathcal{E}_{\mathrm{NA}} = 0$ if and only if the distribution is balanced inside every abelianization fiber:

$$p_e = p_{r^2}, \quad p_r = p_{r^3}, \quad p_s = p_{r^2s}, \quad p_{rs} = p_{r^3s}. \tag{39}$$

Thus $\mathcal{E}_{\mathrm{NA}}$ measures a precise component of the distribution that all one-dimensional group characters miss. It is a standard finite-group Fourier quantity, not a new physical invariant.

4 Two-rate D_4 dynamics and its relaxation spectrum

Consider a continuous-time right random walk generated by nearest quarter-turns and a reflection:

$$(\mathcal{L}f)(g) = \kappa_r\big[f(gr) + f(gr^{-1}) - 2f(g)\big] + \kappa_s\big[f(gs) - f(g)\big], \tag{40}$$

with rates $\kappa_r, \kappa_s \geq 0$. This generator is left-equivariant. It is a minimal symmetric model rather than a universal law.

4.1 Binary-character dynamics

Because each one-dimensional character is an eigenfunction of $\mathcal{L}$,

$$\mathcal{L}\chi_R = -4\kappa_r\chi_R, \tag{41}$$
$$\mathcal{L}\chi_S = -2\kappa_s\chi_S, \tag{42}$$
$$\mathcal{L}\chi_{RS} = -(4\kappa_r + 2\kappa_s)\chi_{RS}. \tag{43}$$

Define the positive relaxation rates

$$\gamma_R = 4\kappa_r, \qquad \gamma_S = 2\kappa_s, \qquad \gamma_{RS} = 4\kappa_r + 2\kappa_s. \tag{44}$$

Then

$$\boxed{\gamma_{RS} = \gamma_R + \gamma_S.} \tag{45}$$

This gives a simple blind-test design: estimate γ_R and γ_S from two binary protocols, freeze the parameters, and predict the third.

4.2 The full spectrum and the missing non-Abelian modes

In the two-dimensional representation, the Fourier symbol is

$$\widehat{\mathcal{L}}(\rho_2) = \kappa_r(R + R^{-1} - 2I) + \kappa_s(S - I) = \begin{pmatrix} -2\kappa_r & 0 \\ 0 & -2\kappa_r - 2\kappa_s \end{pmatrix}. \tag{46}$$

Because ρ_2 occurs twice in the regular representation, each of these two rates has multiplicity two. Hence the complete positive relaxation-rate multiset is

$$\boxed{0, \quad 2\kappa_s, \quad 4\kappa_r, \quad 4\kappa_r + 2\kappa_s, \quad 2\kappa_r\ (\times 2), \quad 2\kappa_r + 2\kappa_s\ (\times 2).} \tag{47}$$

The last four linear degrees of freedom are precisely the two-dimensional non-Abelian sector omitted by homomorphic binary characters.

4.3 Closed-form transition probabilities

Writing $g = r^j s^\chi$ with $j \in \mathbb{Z}_4$ and $\chi \in \mathbb{Z}_2$, define

$$u = e^{-2\kappa_r t}, \qquad v = e^{-2\kappa_s t}. \tag{48}$$

For the symmetric generator in Eq. (40), the rotation displacement probabilities are

$$p_0(t) = \frac{(1+u)^2}{4}, \tag{49}$$

$$p_1(t) = p_3(t) = \frac{1-u^2}{4}, \tag{50}$$

$$p_2(t) = \frac{(1-u)^2}{4}, \tag{51}$$

and the reflection-parity probabilities are

$$q_0(t) = \frac{1+v}{2}, \qquad q_1(t) = \frac{1-v}{2}. \tag{52}$$

Thus

$$P\big[(j,\chi) \to (k,\psi); t\big] = p_{k-j\,(\mathrm{mod}\,4)}(t)\, q_{\psi-\chi\,(\mathrm{mod}\,2)}(t). \tag{53}$$

The factorization is special to the symmetric choice of equal clockwise and counter-clockwise rates.

At short times, the lowest power of t is the shortest word length in the generating set $\{r, r^{-1}, s\}$. For example, quarter-turns and a single reflection are $O(t)$, a half-turn is $O(t^2)$, and a half-turn plus reflection is $O(t^3)$. This provides parameter-independent structural falsification tests for the minimal generator.

5 Information budget and context-rich refinements

The group-theoretic coarse-graining can be compared to an explicitly refined record representation. Let

$$B_3 = \{00, 11, D\}, \qquad D = \{01, 10\}, \tag{54}$$

and introduce a binary fiber label $\eta \in \mathbb{Z}_2$. The six-label set $B_6 = B_3 \times \mathbb{Z}_2$ is a *designed refinement*, not a group quotient. If η is identified operationally with the abelianization fiber coordinate, then B_6 distinguishes e from r^2 and rs from $r^3 s$ but still merges the two mixed coarse classes 10 and 01.

A minimal complete semantic code can therefore be written as (C, η, δ), where $C \in B_3$ and δ is needed only when $C = D$:

$$\begin{aligned} C = 00 : &\quad g = r^{2\eta}, && (55)\\ C = 11 : &\quad g = r^{1+2\eta}s, && (56)\\ C = D,\ \delta = 0 : &\quad g = r^{1+2\eta}, && (57)\\ C = D,\ \delta = 1 : &\quad g = r^{2\eta}s. && (58)\end{aligned}$$

The state count is $2 + 2 + 4 = 8$.

For a uniform distribution on D_4,

$$H(G) = 3 \text{ bits}, \qquad H(B_4) = 2 \text{ bits}, \qquad H(B_3) = 1.5 \text{ bits}. \tag{59}$$

If $B_6 = (B_3, \eta)$ is induced from uniform G, then

$$H(B_6) = 2.5 \text{ bits}, \qquad H(G \mid B_6) = 0.5 \text{ bit}. \tag{60}$$

The remaining half bit is exactly the average cost of δ: it is needed only on the mixed class, which has probability one half.

Now consider an overcomplete source-tagged record space

$$\mathcal{S}_{96} = B_3 \times \mathbb{Z}_2 \times \Lambda^2, \qquad |\Lambda| = 4. \tag{61}$$

Its size alone does not imply additional physical information. Let $(X, Y) \in \Lambda^2$ be source/context labels. Their relevance to instantaneous D_4 recovery is quantified by

$$I(G; X, Y \mid C, \eta). \tag{62}$$

If this conditional mutual information vanishes, increasing the number of source labels cannot recover any additional D_4 state information. Exact recovery requires

$$H(G \mid C, \eta, X, Y) = 0. \tag{63}$$

Thus a large context space should be justified by predictive, control, or reconstruction utility rather than by cardinality alone [4].

6 Coherent recovery of a commutator residue in standard quantum mechanics

6.1 The residue is invisible as a target channel

Using Pauli matrices, the standard two-dimensional representation may be written

$$R = -iY, \qquad S = Z. \tag{64}$$

Then

$$RS = X, \qquad SR = -X, \tag{65}$$

and the group commutator is

$$K = RSR^\dagger S^\dagger = -I. \tag{66}$$

On an isolated target system,

$$\mathrm{Ad}_K(\rho) = K\rho K^\dagger = \rho, \tag{67}$$

so standard process tomography of the target alone reports the identity channel. Likewise $\mathrm{Ad}_{RS} = \mathrm{Ad}_{SR}$ because RS and SR differ only by a global phase.

6.2 A coherent reference converts global phase into relative phase

Prepare a control qubit in $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ and an arbitrary target state ρ. Implement the commutator sequence *conditionally* on the control-$|1\rangle$ branch, gate by gate. The resulting controlled unitary is

$$C(K) = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes K. \tag{68}$$

Because $K = -I$,

$$C(K) = Z_c \otimes I_t. \tag{69}$$

Hence

$$|+\rangle_c \otimes \rho_t \longmapsto |-\rangle_c \otimes \rho_t, \tag{70}$$

with ideal witness

$$\boxed{\langle X_c \rangle_{\text{comm}} = -1.} \tag{71}$$

A matched null sequence such as $RR^\dagger SS^\dagger$ has the same target identity channel but gives

$$\boxed{\langle X_c \rangle_{\text{null}} = +1.} \tag{72}$$

This is a direct instance of phase kickback/Hadamard-test logic [6]. Importantly, compiling the target sequence to a bare identity before introducing the coherent control would erase the witness; the ordered controlled implementation is part of the protocol.

The protocol does not reveal physics beyond quantum theory. The relation $RS = -SR$ is standard Pauli algebra, and the observability of a controlled global phase is textbook quantum mechanics. Its role here is conceptual and operational: a residue that is invisible in the target channel becomes visible relative to a retained reference frame.

7 Relation to quantum instruments and conditional structure

Quantum instruments contain outcome-conditioned completely positive maps whose sum gives a nonselective channel and whose adjoints on the identity give POVM effects [5]. Consequently, discarding the outcome label can erase operational information relevant to adaptive control. Environment-assisted correction provides a related example in which classical information about a branch can restore correctability [8]. Channel holonomy similarly emphasizes gauge and transport structure beyond a single local channel description [7].

The present finite-group construction expresses the same general logical pattern at a simpler algebraic level:

$$\boxed{\text{coarse equivalence does not imply conditional or reference-sensitive equivalence.}} \tag{73}$$

For the D_4 quotient, rs and sr become identical in every one-dimensional character even though they differ in the two-dimensional sector. For the quantum representation, RS and SR induce the same target channel but differ relative to a coherent branch reference. These are standard-theory effects; the value of the framework lies in making the hierarchy of descriptions explicit and testable.

8 A two-stage blind-test protocol

The results suggest a compact validation sequence.

Stage A: abelian binary layer. Implement binary readouts corresponding to χ_R and χ_S, estimate γ_R and γ_S, and freeze all parameters. The third binary readout must satisfy

$$\gamma_{RS}^{\text{pred}} = \gamma_R + \gamma_S. \tag{74}$$

Failure rejects the minimal generator or the assumed readout homomorphisms.

Stage B: non-Abelian reference-sensitive layer. Implement a controlled commutator sequence and a gate-count-matched null sequence. Both must act as the identity channel on the target, while the control-qubit X expectation should have opposite signs in the ideal model. Failure of Stage B after Stage A succeeds would indicate that the system realizes the abelian coarse structure without retaining the coherent two-dimensional representation assumed by the protocol, or that coherence is lost in implementation.

Passing both stages demonstrates a standard D_4-structured implementation; it does not establish a new quantum ontology.

9 Scope, limitations, and novelty boundary

Several boundaries are essential.

1. The abelianization and representation theory of D_4 are standard mathematics. The paper's contribution is the integrated coarse-graining/reconstruction formulation, the explicit probability reconstruction in Eq. (30), the information-budget interpretation, and the linked dynamic and coherent-readout protocols.
2. The phrase "binary coarse-graining" refers here to homomorphic binary observables derived from one-dimensional characters. Arbitrary binary classifiers on the eight states need not be blind to non-Abelian structure.
3. The generator in Eq. (40) is a minimal symmetric random walk. It is not claimed to be a universal microscopic law.
4. The coherent commutator witness is fully explained by standard quantum mechanics. No violation of the Born rule, no hidden-variable completion, and no beyond-CPTP effect is claimed.
5. A source-tagged space such as $\mathcal{S}_{96}$ is a representational choice. Its scientific value depends on conditional mutual information, prediction, reconstruction, or feedback gain, not on the number 96.

These restrictions are not incidental. They separate a falsifiable mathematical/information framework from stronger physical interpretations that would require independent experimental evidence.

10 Conclusion

We have given an exact end-to-end analysis of information loss and recovery for homomorphic binary coarse-graining of the eight-element dihedral group D_4. The natural $8 \to 4$ abelianization deletes one binary fiber coordinate and makes noncommuting paths such as rs and sr indistinguishable. Three nontrivial binary characters reconstruct the full abelian sector, while the unique two-dimensional irreducible representation carries exactly the four remaining linear degrees of freedom. The corresponding Fourier matrix yields an exact inverse reconstruction and a compact non-Abelian residual energy.

For a symmetric two-rate random walk, the same decomposition organizes the entire relaxation spectrum: the binary characters produce three scalar rates with the sum rule $\gamma_{RS} = \gamma_R + \gamma_S$, while the missing doubly degenerate modes belong to the two-dimensional non-Abelian sector. In a standard quantum realization, the commutator $-I$ is invisible as a target-system channel but becomes a measurable relative phase in a coherent reference experiment. Finally, information-theoretic accounting shows that overcomplete source-tagged record spaces should be evaluated by conditional information and control utility rather than state count.

The resulting principle is modest but general: coarse observables can be operationally complete for one task and insufficient for another. Retaining conditional structure, path order, or a coherent reference can restore information that a binary marginal necessarily discards.

Data and code availability

No experimental data are used in this work. The core results are exact algebraic identities. A compact symbolic verification script accompanying this preprint checks the D_4 relations, the two-dimensional generator spectrum, the commutator identity, and the Fourier reconstruction formulas.

Author contributions and AI-tool use

Y. Wang conceived the originating research direction, selected the D_4 coarse-graining problem, and is the sole author responsible for the scientific claims. OpenAI ChatGPT (GPT-5.6 Sol, accessed August 2026) was used under the author's direction for algebraic cross-checking, literature organization, drafting, and document preparation. The author is responsible for independently verifying all claims, references, equations, and any public release.

Competing interests

The author declares no competing interests.

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A Explicit abelianization-fiber table

Full state	Abelian code	Relation class	Fiber sign η
e	00	00	0
r^2	00	00	1
r	10	D	0
r^3	10	D	1
s	01	D	0
r^2s	01	D	1
rs	11	11	0
$r^3s = sr$	11	11	1

The table makes clear that (C, η) is sufficient for the 00 and 11 relation classes, while one additional mixed-class selector δ is required on D.

B Information budget under the uniform prior

For uniform $G \in D_4$, the distributions induced by the hierarchy are

Representation	Outcome probabilities	Entropy (bits)
$G \in D_4$	eight times $1/8$	3
B_4	four times $1/4$	2
B_3	$1/4, 1/4, 1/2$	1.5
$B_6 = (B_3, \eta)$	four times $1/8$, two times $1/4$	2.5

Hence $H(G \mid B_4) = 1$ bit, $H(G \mid B_3) = 1.5$ bits, and $H(G \mid B_6) = 0.5$ bit.

C Short-time word-length scaling

Expanding Eq. (53) for small t gives

$$P(\pm 90^\circ) = \kappa_r t + O(t^2), \tag{75}$$
$$P(\text{reflection}) = \kappa_s t + O(t^2), \tag{76}$$
$$P(180^\circ) = \kappa_r^2 t^2 + O(t^3), \tag{77}$$
$$P(\pm 90^\circ + \text{reflection}) = \kappa_r \kappa_s t^2 + O(t^3), \tag{78}$$
$$P(180^\circ + \text{reflection}) = \kappa_r^2 \kappa_s t^3 + O(t^4). \tag{79}$$

The leading exponent is the shortest path length in the Cayley graph generated by $\{r, r^{-1}, s\}$.